TENSILE TESTING

INTRODUCTION

Stress and strain allows us to relate load to a load bearing area. Stress is the ratio between the amount of force exerted onto an object and the cross-sectional area of the object. Strain is the ratio of the change of length and the original length. With these two components (stress along strain), we can organise these into a curve where we can infer information about the properties of the metal. Such as the metal's hardness (the ability to resist deformation), stiffness (Force required to produce unit of displacement of an object) and toughness (the measure of the amount of energy a materials absorbs as it fractures. We could gather more information about materials through working out 'young's modulus', at certain parts of the curve. My group had demonstrated our knowledge of material properties, in particular of metals and the definitions of stress and strain, by carrying out the tensile testing experiment. In this tested the properties of metals such as: mild steel, aluminium, and alloy. We carried out results to produced multiple stress-strain graphs to infer from. We gathered results for the amount of force exerted after each turn, which aided us towards calculating the stress and strain.

STEP BY STEP

1.First you need to attach the tensile tester in the center of the board making sure all the screws are tight
2.The samples dimensions then need to be measured using a caliper and the cross sectional area needs to be calculated
3.Next the safety guard should be removed to allow the specimen to be mounted between the two chucks and screwed in using an allen key
4.The safety guard should then always be refitted before the test commences
5.The top chuck is connected to the load nut which will pull it upwards and give an extension reading increasing in increments of 1 mm
6.The bottom chuck is connected to two large springs where a dial measures the deflection of the springs as you apply force to the sample
7.Once the new specimen has been fitted the load nut should be turned to align it with one of the marks on its scale so the amount of turns and therefore extension can be kept track of
8.The indicator on the force dial should then be turned to 0
9.The load nut should be turned clockwise and measurements of the extension and force after each after each half turn or turn should be measured
10.This should be continued until the specimen elongates to the point of fracture
11.This test should then be repeated so as to enhance the accuracy of the results

MILD STEEL

Mild steel possesses a combination of strength and flexibility based on our experiments. Our tests showed that mild steel can undergo significant strain before it snaps, making it great for absorbing shocks. This quality is beneficial in construction, where materials often face impact.

Moreover, mild steel's elasticity under relatively high stress, makes it suitable for building structures like bridges and buildings. Its capacity to dampen vibrations also makes it useful in the automotive industry for smoother rides.

However, it's important to note that mild steel has limits, especially in extreme bending situations like crumple zones in car crashes. So as not to pose any unnecessary danger.

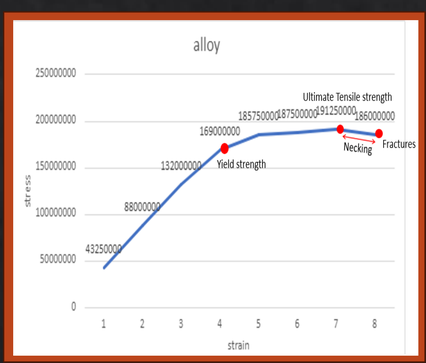
ALLOY

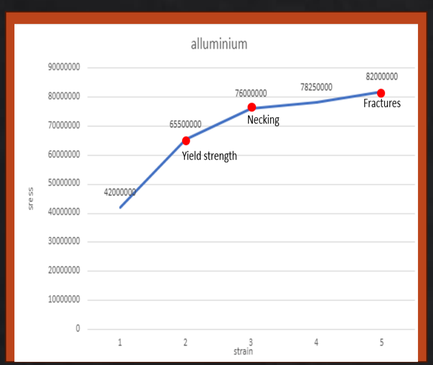
The Alloy, that we used, can go through the most elastic deformation of all three materials we tested

Good for tools and the aerospace industry, because it be able to handle stress, and have a large elastic region to make the tools have a long life.

Because of a lot of elastic strain was observed; this would be a good material, because it would allow for the small amounts of deformation to be negligible, because of its high elastic region.

The total amount of strain that this alloy can withstand is not that much, it means that it might not be good in car crashes, it won't deform very well, if used somewhere such as the crumple zone.





ALUMINIUM

Aluminium is the weakest of all three metals that we tested, as it experiences the least amount of strain and stress before breaking.

This is not bad though because of its very light weight that makes it good in things like the aerospace/automotive industry and consumer goods like phones and laptops

But because of its weak stress-strain properties an alloy is often used when working on planes and cars

Because of its low stress properties it means that its easy to work with making more favorable in some manufacturing

CONCLUSION

To conclude, we found that mild steel was the strongest of the metals, whereas aluminium was the weakest. Mild steel could withstand the most stress and strain than the other materials – it handled the most deformation and force before breaking. Thus, mild steel proved to be the toughest of the metals. Secondly, we found that aluminium was the weakest of the metals as it had entered the plastic region only after 2 turns. Aluminium's lower strength highlights its limitations in terms of load bearing capacity. The properties of this metals would be affective for stronger structures such as in cars and buildings. Finally, the alloy produced the most elastic deformation which is crucial for designing structures which undergo various loads and stress without undergoing permanent deformation. These properties would work out affective for structures such as hammers. We found therefore, that the properties of this metal would be affective for structures such as in planes.

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